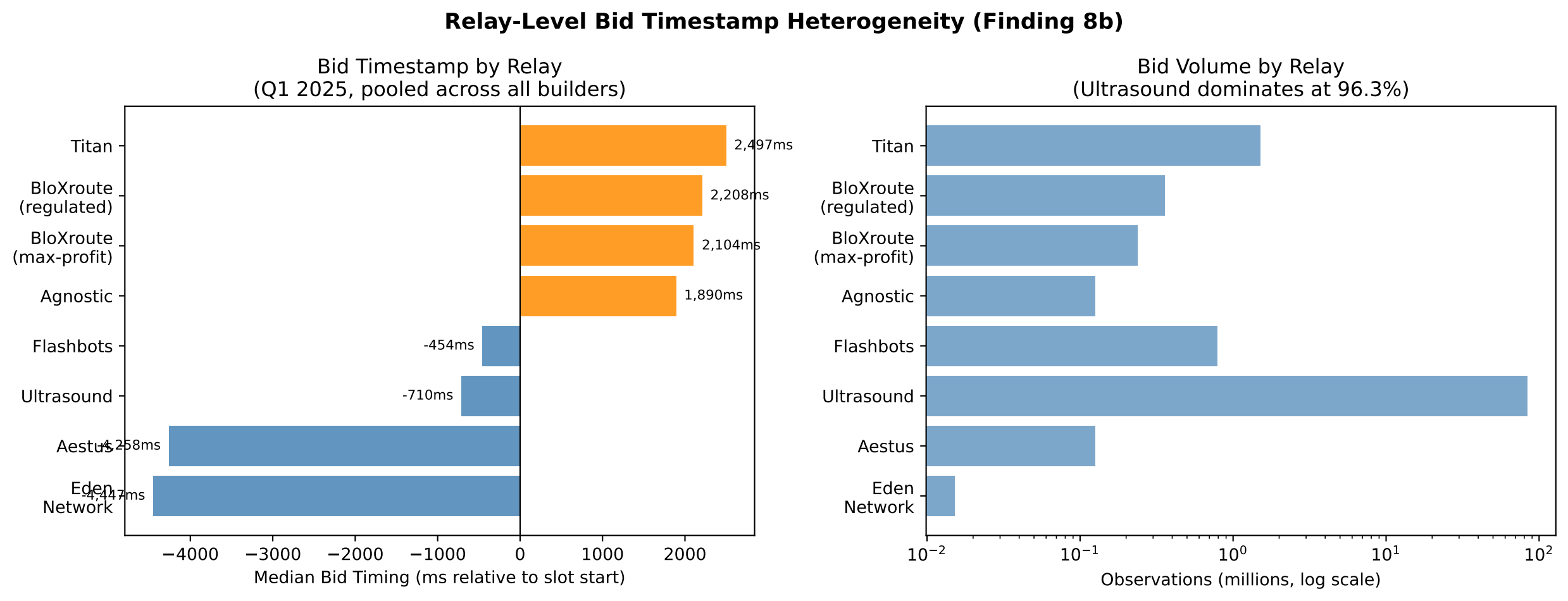
Relay-Level Bid Timestamp Heterogeneity (Finding 8b)
Bid Timestamp by Relay
(Q1 2025, pooled across all builders)
Titan
BloXroute (regulated)
BloXroute (max-profit)
Agnostic
Flashbots
Ultrasound
Aestus
Eden Network
2,497ms
2,208ms
2,104ms
1,890ms
-454ms
-710ms
-4,258ms
-4,447ms
−4000
−3000
−2000
−1000
0
1000
2000
Median Bid Timing (ms relative to slot start)
Bid Volume by Relay
(Ultrasound dominates at 96.3%)
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Observations (millions, log scale)